

Abstract

The aim of this sub-project is to develop an image processing component to support low-cost automated greenhouses. The image processing application must enable the greenhouse system to establish various aspects about the plants from the images taken. These aspects include plant health, plant growth and plant disease identification from the images of different parts of the plants in the greenhouse.

What is the purpose of this section?

- Context and importance of your research.
- Outline of literatur review

Why choose image processing techniques?

- Former methods cost too much time and were expensive + destructive [16]
- Focus on image analysis component [9]
- Review plant phenotyping concepts in relation to image processing and plant breeding, discussing various measurable traits

1 Plant Phenotyping

Plant breeding is defined as the process of inducing desirable inheritable traits in plants through controlled human action [23], which is important in solving the problem of providing an adequate global food supply. The rapid growth of the human population has created a strain on international food security. It is estimated that crop production must increase by 70% by 2050 [6] to meet this demand. Additionally, abiotic and biotic stressors limit cultivar biomass and yield [11]. Therefore, plant breeders need effective techniques to produce germplasm for cultivars that maximize yield and biomass, in various adverse environmental conditions. In the last few decades, image processing techniques in plant phenotyping have been an effective tool for plant breeding [3].

1.1 What Is Plant Phenotyping?

Plant phenotyping is the evaluation of phenotypes in plants [10], where phenotypes are primitive or complex traits of an organism that can be observed or quantified and evaluated numerically [14]. Phenotypes can alternatively be seen as combination of all morphological, physiological, chemical, and behavioral aspects that represent the individual plant [12, 22, 19].

1.2 Plant Phenotyping As Method For Improving Plant Traits

Quantitative data produced by plant phenotyping is most useful in analyzing plant-environment interactions in plant breeding [10] for the development of germplasm with improved plant traits [19]. The data produced by phenotyping is useful as a proxy for assessing a plant's performance in relation to its environment. For example, stressors or disturbances of plants are induced by environmental conditions that organisms are placed in, and these stressors induce symptoms which can be detected and evaluated using phenotyping. This indicates phenotypical traits of a plant have complex dynamic feedback loop with their environment [33], and therefore phenotypical traits serve as proxies in determining whether certain genotypes have favourable traits in particular environmental conditions.

1.3 Image-Obtainable Phenotypical Traits As Indicators Of Plant Health, Growth, And Stress

A phenotypical trait can be classified as a morphological, physiological, or temporal trait. Additionally the literature indicates, morphological and physiological traits are classifiable as holistic-based or component-based [7]. Holistic properties consider the plant as a whole, whereas component-based features look at particular organs like the fruit, flower, leaves, stem etc. of the organism.

Leaf-derived traits are one of the most commonly used types of plant status indicators, because leaves rapidly respond to environmental stimuli, and robustly indicate various factors which determine plant health, growth and stress. [37]. Traits of leaves can monitor and predict a host of plant health, growth and stress factors. For example, leaf area predicts plant productivity, because reduced leaf area results in decreased chlorophyll content, and consequently less photosynthesis. Therefore leaf area is a good proxy measure for evaluating plant growth, and has been used to monitor and estimate crop yield, and health [38]. Various other morphological traits have also been utilized in plant phenotyping. These include leaf number and inclination angle for monitoring plant growth and development [26, 37], and leaf thickness for measuring abiotic stressors like water scarcity [2]. Additionally, physiological traits of leaves exist that indicate various factors of plant status like stomatal conductance for the detection of biotic stressors like pathogens [13], leaf water content for drought stress [34].

Although leaves play an important role in evaluating plant growth, health and resilience to stress, these evaluations cannot be scaled to the whole plant [37]. Rather, various other parts of the plant should be phenotyped to comprehensively understand the processes that govern plant status. The literature is rife with such examples that study traits of roots [5], stems [8], seeds [36], and fruit [1].

2 The Relationship Between Image Processing And Plant Phenotyping

Plant phenotyping methods with image processing are non-destructive, amenable to automation, and readily scalable to high throughput scenarios in large samples. This is in stark contrast to traditional phenotyping methods which are prone to human error, extremely time-consuming, costly, and are difficult to perform from start to end of a plant's ontogenesis [4, 37, 7].

What is the purpose of this section?

- Discuss image processing in the context of plant phenotyping
- Discuss image acquisition techniques
- Discuss image processing techniques used in plant phenotyping

How does image processing relate to plant phenotyping?

Fill out this section

The conceptual pipeline for image processing techniques in plant phenotyping

This paragraph should discuss how image processing is applied. I.e it should map the process from image acquisition to image processing to data generation/analysis

Foundations of image acquisition in plant phenotyping

- measure a phenotype quantitatively through the interaction between light and plants such as reflected photons, absorbed photons, or transmitted photons. Each component of plant cells and tissues has wavelength-specific absorbance, reflectance, and transmittance properties [19]
- For example, chlorophyll absorbs photons primarily in the blue and red spectral region of visible light [19]
- Imaging at different wavelengths is used for different aspects of plant phenotyping [19].
- Imaging techniques are very helpful for detecting these properties, especially for those that cannot be seen by the naked eye [19]

Image acquisition methodologies in plant phenotyping

Thermal imaging sensors

- Conceptual basis of this technique [25]
- Have been used to measure plant water status.
- Used in Irrigation scheduling.
- By inferring transpiration rate of leaves which is governed by stomatal conductance.
- Stomatal conductance can indicate drought, pathogens, and also soil salinity.
- Have also been used for yield predictions [13]

Hyperspectral sensors

- Based on the principles of spectroscopy. Can detect biochemical, morphological and physiological traits of plants. Has been used to detect drought stress. Also used to detect water stress, and pathogens. Different reflectance can tell us about different things. Certain band tells us about water stress, some can tell us about chlorophyll content. [29].
- Conceptual basis and applications [28]
- Crop yield
- Biotic and abiotic factors

Tomography

- three-dimensional visualization of plant internal structures, root systems, and physiological processes
- Based on principles of x-rays, moving sensor takes planar cross sections to 3D visualize plant
- Drought stress [32]
- Yield Estimations
- has references for usages [24]

Fluorescence sensors

- Conceptual basis of this technique [30]
- Predict metabolism using photosynthesis which tells us about growth status

- Any stress or disturbance can indirectly or directly affect plant metabolism, so chlorophyll content can be used as a measure to detect these things like pathogens, light shortage, nitrogen deficiency, water stress etc [20]
- In optimal conditions, very little heat is emitted in photosynthesis so in certain near infrared emission should be low
- Actinic light involved here?

RGB Camera

- Mimic human vision
- measure plant shoot biomass (yield potential) [19]
- leaf senescence (tissue resistance to salt)

What is the purpose of this section?

- Discuss how image processing and plant phenotyping are applied in low-cost automated greenhouses
- What methods, limitations and advances have there been.
- What does a typical image preprocessing pipeline look like?

Basis for future in low-cost imaging analysis

The basis for the points below is from [nabwireReviewApplicationArtificial2021]

- Existing phenotyping systems are commercially developed and under patent, so are not modifiable or readily available [17]
- RGB, and hyperspectral sensors most commonly used as they are affordable
- Data collection and analysis tools are not generalizable or standardised, so many custom flows are made
- Indicating the need for standardised, extendable image analysis pipelines using readily available open-source software and hardware
- Recent advances show the use of cheap hardware like raspberry pi, and affordable RGB sensors [31]
- However these sensors don't have a lot of compute power, so analysis and preprocessing is done remotely
- Cheap sensors but are cheaper and faster
- Cloud architectures exist, but with amount of data generated this cost can be prohibitive in terms of processing cost [bauckhageDataMiningPattern2013]. Cyberinfrastructure expensive ref (95,96)
- Moreover, existing systems are not designed for multiple genotypes so their applicability is limited, suggesting need for more generalizable image analysis
- Deep learning and machine learning can be used to automate image analysis, and reduce costs
- Availability of large labelled data is bottleneck
- To use on cheap hardware you also need to optimize
- Therefore low complexity efficient algorithms for automated image preprocessing and image analysis is needed
- Using multimodal segmentation is also another angle

- ref[36], one of the advantages of using ML approaches in plant phenotyping is their ability to search large datasets and discover patterns by simultaneously looking at a combination of features (compared to analyzing each feature separately). This was previously a challenge because of the high dimensionality of plant images and their large quantity, making them difficult to analyze through traditional processing methods (ref 37)
- machine learning achieved using supervised learning, so we are not able to detect novel or unexpected traits using unsupervised learning (ref 40)
- traditional methods in image processing not generalizable, in pathogen detection and require strict conditions and custom workflows to work [4]
- Current research Focus on image sensing, rather on computation which is new bottleneck. Imaging technology has advanced, but there is lack of robust analysis and computation tools. [bauckhageDataMiningPattern2013]
- Sensors generate huge amount of data, therefore algorithms need to be designed to deal with Terabytes of data. Furthermore, low cost computing units have limited bandwidth, battery and sensor capabilities indicating that algorithms need to take this in to account. Low complexity and efficient algorithms that can run on resource constrained tech is needed [bauckhageDataMiningPattern2013]
- Mobile applications can be used and run processing without need for external network, cyberinfrastructure. However there are not many applications available for this [muller-linowPlantScreenMobile2019]. Image analysis tools are often too expensive to run on these devices due to the constrained resources, however they are much cheaper and therefore can be exploited. In particular low complexity efficient systems, which generalize to various genotypes, and phenotypic measurements are needed. In particular the development of efficient low resource intensive machine modelling and deep learning techniques can be made.
- preprocessing technique choice is also needed, for example image segmentation methods can be costly, and a suitable method depends on various factors. Certain methods have cheaper cost, but less accuracy [muller-linowPlantScreenMobile2019]. In developing low-cost image analysis and preprocessing tools for phenotyping it is necessary to weight the cost savings versus accuracy and precision. But bespoke image processing techniques don't lend well to generalizability and extendability.
- cost-effective solutions require carefully choosing sensors and data processing techniques to be most cost-effective [37]
- automated systems not built for general solutions but for specific species of plants [10]

Current applications

- Often combined with machine learning algorithms
- Cheap Raspberry PI with RGB camera [31]
- Autonomous robots with sensors and cameras [27]
- use off the shelf commercial software like Microsoft Kinect [18]

What affects a low-cost automated greenhouse cost

Points from this paper [15]

- Image preprocessing pipeline
- open source software
- hardware
- amount of labour and time needed for analysis
- energy cost of technology
- amount of data generated, where is it stored, how frequently? [9]

Challenges of low-cost automated greenhouses

- Many phenotyping systems existing are commercially developed under patent so hardware and software can not be modified. [17]
- Current applications use moving conveyor belt (benchtop) [17]
- or conveyer belt [17]
- or moving robotic arm to move sensors around to get high-throughput measurements.
- Robot solutions not cheap for farmers [27]
- large amounts of data need to be collected, and processed. Existing phenotyping platforms exist, but they are expensive and have restricted running environments.
- Many systems developed for laboratory systems, but not extended for real applications. Standardised off the shelf equipment, open-source software allow modular and extensible plus affordable solutions [21]
- Custom data collection schemes need to be made very little standards
- lack of robust methods to analyse plant images
- low cost moving track systems limited in phenotypical, genotype and management types are supported [35] suggesting need for low-cost systems that can support a wider variety of genotypes and phenotypical traits.

Fill out the conclusion section.

References

- [1] Alebel Mekuriaw Abebe et al. “Image-Based High-Throughput Phenotyping in Horticultural Crops”. In: *Plants* 12.10 (10 Jan. 2023), p. 2061. ISSN: 2223-7747. DOI: 10.3390/plants12102061. URL: <https://www.mdpi.com/2223-7747/12/10/2061> (visited on 05/25/2025).
- [2] Amin Afzal, Sjoerd W. Duiker, and John E. Watson. “Leaf Thickness to Predict Plant Water Status”. In: *Biosystems Engineering* 156 (Apr. 1, 2017), pp. 148–156. ISSN: 1537-5110. DOI: 10.1016/j.biosystemseng.2017.01.011. URL: <http://www.scopus.com/inward/record.url?scp=85013072400&partnerID=8YFLogxK> (visited on 05/25/2025).
- [3] Achala Anand, Madhumitha Subramanian, and Debasish Kar. “Breeding Techniques to Dispense Higher Genetic Gains”. In: *Frontiers in Plant Science* 13 (Jan. 19, 2023). ISSN: 1664-462X. DOI: 10.3389/fpls.2022.1076094. URL: <https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2022.1076094/full> (visited on 05/16/2025).
- [4] Jayme Garcia Arnal Barbedo. “Digital Image Processing Techniques for Detecting, Quantifying and Classifying Plant Diseases”. In: *SpringerPlus* 2.1 (Dec. 7, 2013), p. 660. ISSN: 2193-1801. DOI: 10.1186/2193-1801-2-660. URL: <https://doi.org/10.1186/2193-1801-2-660> (visited on 05/11/2025).
- [5] Gernot Bodner et al. “Hyperspectral Imaging: A Novel Approach for Plant Root Phenotyping”. In: *Plant Methods* 14.1 (Oct. 3, 2018), p. 84. ISSN: 1746-4811. DOI: 10.1186/s13007-018-0352-1. URL: <https://doi.org/10.1186/s13007-018-0352-1> (visited on 05/25/2025).
- [6] M. Carvajal-Yepes et al. “A Global Surveillance System for Crop Diseases”. In: *Science* 364.6447 (June 28, 2019), pp. 1237–1239. ISSN: 0036-8075, 1095-9203. DOI: 10.1126/science.aaw1572. URL: <https://www.science.org/doi/10.1126/science.aaw1572> (visited on 05/21/2025).
- [7] Sruti Das Choudhury, Ashok Samal, and Tala Awada. “Leveraging Image Analysis for High-Throughput Plant Phenotyping”. In: *Frontiers in Plant Science* 10 (Apr. 24, 2019). ISSN: 1664-462X. DOI: 10.3389/fpls.2019.00508. URL: <https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2019.00508/full> (visited on 05/21/2025).
- [8] Sruti Das Choudhury et al. “Automated Stem Angle Determination for Temporal Plant Phenotyping Analysis”. In: *Proceedings of the IEEE International Conference on Computer Vision Workshops*. 2017, pp. 2022–2029. URL: https://openaccess.thecvf.com/content_ICCV_2017_workshops/w29/html/Choudhury_Automated_Stem_Angle_ICCV_2017_paper.html (visited on 05/25/2025).

- [9] Jianjun Du et al. “Greenhouse-Based Vegetable High-Throughput Phenotyping Platform and Trait Evaluation for Large-Scale Lettuces”. In: *Computers and Electronics in Agriculture* 186 (July 2021), p. 106193. ISSN: 01681699. DOI: 10.1016/j.compag.2021.106193. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0168169921002106> (visited on 05/24/2025).
- [10] Fabio Fiorani and Ulrich Schurr. “Future Scenarios for Plant Phenotyping”. In: *Annual Review of Plant Biology* 64.1 (Apr. 29, 2013), pp. 267–291. ISSN: 1543-5008, 1545-2123. DOI: 10.1146/annurev-arplant-050312-120137. URL: <https://www.annualreviews.org/doi/10.1146/annurev-arplant-050312-120137> (visited on 05/14/2025).
- [11] J.P. Grime. “Evidence for the Existence of Three Primary Strategies in Plants and Its Relevance to Ecological and Evolutionary Theory”. In: *The American Naturalist* 111.982 (1997), p. 1169. DOI: 10.1086/283244. URL: <https://www.journals.uchicago.edu/doi/abs/10.1086/283244> (visited on 05/19/2025).
- [12] Qiaosheng Guo and Zaibiao Zhu. “Phenotyping of Plants”. In: *Encyclopedia of Analytical Chemistry*. John Wiley & Sons, Ltd, 2014, pp. 1–15. ISBN: 978-0-470-02731-8. DOI: 10.1002/9780470027318.a9934. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/9780470027318.a9934> (visited on 05/14/2025).
- [13] Roselyne Ishimwe, K. Abutaleb, and F. Ahmed. “Applications of Thermal Imaging in Agriculture—A Review”. In: *Advances in Remote Sensing* 03.03 (2014), pp. 128–140. ISSN: 2169-267X, 2169-2688. DOI: 10.4236/ars.2014.33011. URL: <http://www.scirp.org/journal/doi.aspx?DOI=10.4236/ars.2014.33011> (visited on 05/19/2025).
- [14] W. Johannsen. “The Genotype Conception of Heredity”. In: *The American Naturalist* 45.531 (Mar. 1911), pp. 129–159. ISSN: 0003-0147, 1537-5323. DOI: 10.1086/279202. URL: <https://www.journals.uchicago.edu/doi/10.1086/279202> (visited on 05/14/2025).
- [15] Yonghyun Kim, Jinyoung Y. Barnaby, and Scott E. Warnke. “Development of a Low-Cost Automated Greenhouse Imaging System with Machine Learning-Based Processing for Evaluating Genetic Performance of Drought Tolerance in a Bentgrass Hybrid Population”. In: *Computers and Electronics in Agriculture* 224 (Sept. 1, 2024), p. 108896. ISSN: 0168-1699. DOI: 10.1016/j.compag.2024.108896. URL: <https://www.sciencedirect.com/science/article/pii/S0168169924002874> (visited on 05/24/2025).
- [16] R. H. Laxman et al. “Non-Invasive Quantification of Tomato (*Solanum Lycopersicum* L.) Plant Biomass through Digital Imaging Using Phenomics Platform”. In: *Indian Journal of Plant Physiology* 23.2 (June 1, 2018), pp. 369–375. ISSN: 0974-0252. DOI: 10.1007/s40502-018-0374-8. URL: <https://doi.org/10.1007/s40502-018-0374-8> (visited on 05/24/2025).

- [17] Daoliang Li et al. “High-Throughput Plant Phenotyping Platform (HT3P) as a Novel Tool for Estimating Agronomic Traits From the Lab to the Field”. In: *Frontiers in Bioengineering and Biotechnology* 8 (Jan. 13, 2021), p. 623705. ISSN: 2296-4185. DOI: 10.3389/fbioe.2020.623705. URL: <https://www.frontiersin.org/articles/10.3389/fbioe.2020.623705/full> (visited on 05/24/2025).
- [18] Dawei Li et al. “Digitization and Visualization of Greenhouse Tomato Plants in Indoor Environments”. In: *Sensors* 15.2 (2 Feb. 2015), pp. 4019–4051. ISSN: 1424-8220. DOI: 10.3390/s150204019. URL: <https://www.mdpi.com/1424-8220/15/2/4019> (visited on 05/24/2025).
- [19] Lei Li, Qin Zhang, and Danfeng Huang. “A Review of Imaging Techniques for Plant Phenotyping”. In: *Sensors (Basel, Switzerland)* 14.11 (Oct. 24, 2014), pp. 20078–20111. ISSN: 1424-8220. DOI: 10.3390/s141120078. PMID: 25347588. URL: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4279472/> (visited on 04/20/2025).
- [20] Hartmut K. Lichtenthaler and Ursula and Rinderle. “The Role of Chlorophyll Fluorescence in The Detection of Stress Conditions in Plants”. In: *C R C Critical Reviews in Analytical Chemistry* 19 (supl Jan. 1, 1988), S29–S85. ISSN: 0007-8980. DOI: 10.1080/15476510.1988.10401466. URL: <https://doi.org/10.1080/15476510.1988.10401466> (visited on 05/22/2025).
- [21] Massimo Minervini, Hanno Scharr, and Sotirios A. Tsafaris. “Image Analysis: The New Bottleneck in Plant Phenotyping”. In: *IEEE Signal Processing Magazine* 32.4 (July 2015), pp. 126–131. ISSN: 1053-5888, 1558-0792. DOI: 10.1109/MSP.2015.2405111. URL: <https://ieeexplore.ieee.org/document/7123050/> (visited on 05/24/2025).
- [22] Semira Ebrahim Mohammed. “Review on Plant Phenotyping”. In: (). URL: <https://www.abrinternationaljournal.org/abstract/review-on-plant-phenotyping-1102322.html> (visited on 05/05/2025).
- [23] Thomas J. Orton. “Introduction”. In: *Horticultural Plant Breeding*. Elsevier, 2020, pp. 3–7. ISBN: 978-0-12-815396-3. DOI: 10.1016/B978-0-12-815396-3.09986-8. URL: <https://linkinghub.elsevier.com/retrieve/pii/B9780128153963099868> (visited on 05/16/2025).
- [24] Stefan Paulus. “Measuring Crops in 3D: Using Geometry for Plant Phenotyping”. In: *Plant Methods* 15.1 (Sept. 3, 2019), p. 103. ISSN: 1746-4811. DOI: 10.1186/s13007-019-0490-0. URL: <https://doi.org/10.1186/s13007-019-0490-0> (visited on 05/23/2025).
- [25] Anupma Prakash. “THERMAL REMOTE SENSING: CONCEPTS, ISSUES AND APPLICATIONS”. In: *International Archives of Photogrammetry and Remote Sensing* 33 (2000), pp. 239–243.

- [26] J. Praveen Kumar and S. Domnic. “Rosette Plant Segmentation with Leaf Count Using Orthogonal Transform and Deep Convolutional Neural Network”. In: *Machine Vision and Applications* 31.1–2 (Feb. 2020), p. 6. ISSN: 0932-8092, 1432-1769. DOI: 10.1007/s00138-019-01056-2. URL: <http://link.springer.com/10.1007/s00138-019-01056-2> (visited on 05/25/2025).
- [27] Amine Saddik et al. “Mapping Agricultural Soil in Greenhouse Using an Autonomous Low-Cost Robot and Precise Monitoring”. In: *Sustainability* 14.23 (23 Jan. 2022), p. 15539. ISSN: 2071-1050. DOI: 10.3390/su142315539. URL: <https://www.mdpi.com/2071-1050/14/23/15539> (visited on 05/24/2025).
- [28] Rijad Sarić et al. “Applications of Hyperspectral Imaging in Plant Phenotyping”. In: *Trends in Plant Science* 27.3 (Mar. 1, 2022), pp. 301–315. ISSN: 1360-1385. DOI: 10.1016/j.tplants.2021.12.003. URL: <https://www.sciencedirect.com/science/article/pii/S1360138521003460> (visited on 05/22/2025).
- [29] Michèle R. Slaton, E. Raymond Hunt Jr., and William K. Smith. “Estimating Near-Infrared Leaf Reflectance from Leaf Structural Characteristics”. In: *American Journal of Botany* 88.2 (2001), pp. 278–284. ISSN: 1537-2197. DOI: 10.2307/2657019. URL: <https://onlinelibrary.wiley.com/doi/abs/10.2307/2657019> (visited on 05/22/2025).
- [30] Tingting Tan et al. “Far-Red Light: A Regulator of Plant Morphology and Photosynthetic Capacity”. In: *The Crop Journal* 10.2 (Apr. 2022), pp. 300–309. ISSN: 2214-5141. DOI: 10.1016/j.cj.2021.06.007. URL: <https://linkinghub.elsevier.com/retrieve/pii/S2214514121001501> (visited on 05/22/2025).
- [31] Jose C. Tovar et al. “Raspberry Pi-Powered Imaging for Plant Phenotyping”. In: *Applications in Plant Sciences* 6.3 (2018), e1031. ISSN: 2168-0450. DOI: 10.1002/aps3.1031. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/aps3.1031> (visited on 05/24/2025).
- [32] Vincent Vadez et al. “LeasyScan: A Novel Concept Combining 3D Imaging and Lysimetry for High-Throughput Phenotyping of Traits Controlling Plant Water Budget”. In: *Journal of Experimental Botany* 66.18 (Sept. 2015), pp. 5581–5593. ISSN: 0022-0957, 1460-2431. DOI: 10.1093/jxb/erv251. URL: <https://academic.oup.com/jxb/article-lookup/doi/10.1093/jxb/erv251> (visited on 05/23/2025).
- [33] Achim Walter, Frank Liebisch, and Andreas Hund. “Plant Phenotyping: From Bean Weighing to Image Analysis”. In: *Plant Methods* 11.1 (Mar. 4, 2015), p. 14. ISSN: 1746-4811. DOI: 10.1186/s13007-015-0056-8. URL: <https://doi.org/10.1186/s13007-015-0056-8> (visited on 05/05/2025).

- [34] Kenta Watanabe et al. “Fundamental Study on Water Stress Detection in Sugarcane Using Thermal Image Combined with Photosynthesis Measurement Under a Greenhouse Condition”. In: *Sugar Tech* 24.5 (Oct. 1, 2022), pp. 1382–1390. ISSN: 0974-0740. DOI: 10.1007/s12355-021-01087-y. URL: <https://doi.org/10.1007/s12355-021-01087-y> (visited on 05/25/2025).
- [35] Rafael Massahiro Yassue et al. “A Low-cost Greenhouse-based High-throughput Phenotyping Platform for Genetic Studies: A Case Study in Maize under Inoculation with Plant Growth-promoting Bacteria”. In: *The Plant Phenome Journal* 5.1 (Jan. 2022), e20043. ISSN: 2578-2703, 2578-2703. DOI: 10.1002/ppj2.20043. URL: <https://access.onlinelibrary.wiley.com/doi/10.1002/ppj2.20043> (visited on 05/24/2025).
- [36] Chongyuan Zhang et al. “High-Throughput Phenotyping of Seed/Seedling Evaluation Using Digital Image Analysis”. In: *Agronomy* 8.5 (5 May 2018), p. 63. ISSN: 2073-4395. DOI: 10.3390/agronomy8050063. URL: <https://www.mdpi.com/2073-4395/8/5/63> (visited on 05/25/2025).
- [37] Huichun Zhang et al. “High-Throughput Phenotyping of Plant Leaf Morphological, Physiological, and Biochemical Traits on Multiple Scales Using Optical Sensing”. In: *The Crop Journal* 11.5 (Oct. 1, 2023), pp. 1303–1318. ISSN: 2214-5141. DOI: 10.1016/j.cj.2023.04.014. URL: <https://www.sciencedirect.com/science/article/pii/S2214514123000740> (visited on 05/20/2025).
- [38] Juanjuan Zhang et al. “Leaf Area Index Estimation Model for UAV Image Hyperspectral Data Based on Wavelength Variable Selection and Machine Learning Methods”. In: *Plant Methods* 17.1 (May 3, 2021), p. 49. ISSN: 1746-4811. DOI: 10.1186/s13007-021-00750-5. URL: <https://doi.org/10.1186/s13007-021-00750-5> (visited on 05/25/2025).